

Bond lengths from spectroscopy, from theory, and from both together

A test on three monofluorinated molecules with published structures

Short version. Neither source of information wins outright. Quantum chemistry alone gets the overall shape of a molecule roughly right but puts the carbon–fluorine bond 27 to 31 thousandths of an Ångström too long in every one of the three molecules tested, always in the same direction — a systematic bias, not scatter. Rotational spectroscopy alone corrects that bias well when there is enough data, and is erratic when there is not: on a single isotopologue it barely improves on the geometry it started from. Combining the two beats theory in all 6 of the 6 molecule-and-data-level combinations tested, both on overall bond accuracy and on the C–F bond in particular — but only if they are merged the right way, and the right way depends on how much data there is. Merged the wrong way, the hybrid is worse than plain theory in 2 of the 6.

1. What was being tested

A molecule's structure — which atoms are how far apart, and at what angles — can be worked out two quite different ways.

From theory. Solve the equations of quantum mechanics for the molecule's electrons and find the atomic arrangement with the lowest energy. This needs no experiment at all, but the answer is only as good as the approximations used, and cheap approximations have systematic biases.

From experiment. Measure how fast the molecule tumbles in the gas phase. Microwave spectroscopy gives three numbers per molecule — the rotational constants A, B and C — which are fixed by how the mass is distributed. Swap one atom for a heavier isotope and the mass distribution changes in a known way, giving three more numbers. Enough isotopic variants and the structure can be solved for.

The catch with the experimental route is that three numbers per variant is not many. A seven-atom molecule has fifteen independent structural parameters; one measured species gives three numbers. The problem is underdetermined, and there is a whole family of structures that fit the data equally well.

The hybrid idea. Use the data for the directions the data can see, and use theory for the rest. This report asks whether that actually works — specifically, whether the combination beats theory on its own, which is the practical question, because theory on its own is always available and costs nothing but computer time.

Fluorinated molecules make this sharper than usual. Fluorine has only one stable isotope, so no isotopic substitution ever moves the fluorine atom. Its position has to be inferred indirectly, from the requirement that the molecule's centre of mass sits where the measurements say it does, once every other atom has been located. That makes the C–F bond the hardest parameter in the molecule and a good stress test.

2. The four methods compared

Method	What it uses	What it does
theory	no experimental data	Minimises the quantum-chemical energy. The answer depends only on the molecule, not on any measurement.
experiment	rotational constants only	Adjusts the geometry until the calculated rotational constants match the measured ones. Directions the data cannot see are left to drift.
hybrid (split)	both	Splits the parameter space in two using an SVD of the sensitivity matrix. Data owns every direction it can see; theory owns the remainder outright.

Method	What it uses	What it does
hybrid (joint)	both	Leaves every direction contested and weights the two sources by how well each knows that direction. Theory's weight is set by the distance over which it is trusted, here 0.005 Å.

The two hybrids differ only in how the two sources are merged. Both were run for every molecule and every data level, because the earlier work in this repository found that which one is better flips depending on how much data there is — and that flip turns out to reproduce here.

3. The molecules, and why these three

Each molecule needed a published experimental structure *and* the rotational constants from the same study, so that the structure being aimed at and the data being fed in come from the same experiment. The two must also be mutually consistent: feeding the published geometry back through the rotational-constant calculation should reproduce the published constants to within about 1%, which is the size of the unavoidable zero-point-motion offset. A larger gap means one of the two numbers is wrong, and the comparison would be measuring that error rather than anything about the methods.

Molecule	Formula	Atoms	Structural parameters	Isotopic variants	Self-consistency
Vinyl fluoride	CH ₂ =CHF	6	12	6	1.06%
Acetyl fluoride	CH ₃ COF	7	15	6	0.32%
Fluoroethane	CH ₃ CH ₂ F	8	18	6	0.64%

Sources: **Vinyl fluoride** — Hayashi & Inagusa, J. Mol. Spectrosc. 138, 135 (1989); NIST CCCBDB; **Acetyl fluoride** — Pierce & Krisher, J. Chem. Phys. 31, 875 (1959); NIST CCCBDB; **Fluoroethane** — Kraitchman & Dailey (1955); NIST CCCBDB.

A fourth candidate, formyl fluoride (HCOF), was checked and dropped. Its tabulated geometry and its tabulated constants disagree by 6.2% — far beyond any zero-point effect — so at least one of them is wrong and it cannot serve as a ground truth. It is mentioned here because the selection test is the part of the setup most likely to be skipped, and skipping it would have silently poisoned a quarter of the results.

4. How the test was run

Level of theory

Hartree–Fock with the 6-31G basis set, through PySCF, for every molecule and every method — including the quantum half of both hybrids. This is deliberately a modest level of theory. The point is not to produce the best possible structures; it is to compare three ways of using the same quantum surface, and holding the surface fixed is what makes the comparison mean anything. It is also the same level used on fluorobenzene earlier in this project, so those numbers are comparable with these.

Starting geometry

Every method starts from the same deliberately wrong geometry: the published structure with random noise of 0.03 Å added to every coordinate (one fixed random seed, so the runs are reproducible). Without this, a method could look good simply by not moving. The starting error is reported in every table as the do-nothing baseline that any method has to beat.

The rotational constants

The parent molecule's constants are the published measured values. The constants for the isotopically substituted variants are *derived* from the published structure rather than measured, because the measured values, while they exist in the original papers, are not in any source reachable from this environment. They are not simply the rigid values of the reference geometry: each is scaled by the ratio the real parent

molecule shows between its measured constants and the rigid value of the published structure. That reproduces the offset real ground-state constants carry from zero-point vibration, so the data is not exactly consistent with the answer and a fit cannot recover it for free.

This is the weakest link in the setup and is worth being blunt about. It means the substituted-species data is more internally consistent than real measurements would be, which flatters the experiment-only and hybrid legs at the higher data level. It does not affect the theory leg at all, and it does not affect the parent-only results, which use nothing but published measured numbers.

Two data levels

Data level	Species	Numbers measured	Compared with
Vinyl fluoride — parent only	1	3	12 parameters
Vinyl fluoride — all species	6	18	12 parameters
Acetyl fluoride — parent only	1	3	15 parameters
Acetyl fluoride — all species	6	18	15 parameters
Fluoroethane — parent only	1	3	18 parameters
Fluoroethane — all species	6	18	18 parameters

The first level is the realistic hard case: one measured species, three numbers, far fewer than the molecule has parameters. The second is the comfortable case where the count of measurements reaches or passes the count of parameters. The gap between them is where the interesting behaviour lives.

How error is measured

Each fitted geometry is reduced to chemically meaningful numbers — bond lengths and bond angles, with symmetry-equivalent ones averaged — and compared with the same numbers from the published structure. Bond errors are quoted in milli-Ångström (mÅ, thousandths of an Ångström); for scale, a carbon–carbon bond is about 1500 mÅ long and a good structure determination is accurate to a few mÅ. Angle errors are in degrees. The C–F bond error is reported separately and with its sign, because it is the parameter the whole exercise is about.

5. Results

Three tables summarise all 21 fits. The first is the headline number — overall bond-length accuracy. The second isolates the carbon–fluorine bond. The third covers angles.

5.1 Overall bond-length error (root-mean-square)

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	22.1	19.0	21.1	20.5	18.3
	all species	22.1	19.0	7.3	7.3	8.4
Acetyl fluoride	parent only	35.5	15.8	34.5	27.7	15.4
	all species	35.5	15.8	4.9	4.1	13.6
Fluoroethane	parent only	22.8	15.0	14.2	13.0	11.9
	all species	22.8	15.0	4.1	3.0	5.4

Lower is better; the *start* column is the deliberately wrong geometry every method began from, so any method above its own start column has made things worse. All values in mÅ; the best of the four methods in each row is highlighted.

5.2 The carbon–fluorine bond, signed

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	+10.1	+30.7	-2.6	-2.5	+25.1
	all species	+10.1	+30.7	+10.8	+10.9	+10.5
Acetyl fluoride	parent only	+51.4	+28.8	+36.2	+27.4	+24.5
	all species	+51.4	+28.8	+2.7	+4.7	+22.1
Fluoroethane	parent only	+7.3	+26.9	-0.8	+4.3	+20.4
	all species	+7.3	+26.9	+4.3	+3.3	+8.2

A positive number means the bond came out too long. Theory is positive and large in all three molecules. All values in mÅ; the best of the four methods in each row is highlighted.

5.3 Bond-angle error (root-mean-square)

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	3.37	1.60	1.94	1.77	1.53
	all species	3.37	1.60	2.03	2.02	0.43
Acetyl fluoride	parent only	0.72	1.19	1.19	1.16	1.27
	all species	0.72	1.19	0.22	0.21	1.29
Fluoroethane	parent only	2.04	0.63	0.89	0.76	0.88
	all species	2.04	0.63	0.49	0.25	0.61

Angles behave far less consistently than bond lengths: no method wins across the board, and theory is competitive here in a way it never is for the C–F bond. All values in degrees; the best of the four methods in each row is highlighted.

6. Molecule by molecule

The tables below give every bond length and bond angle, for the published structure and for each method, at both data levels. Symmetry-equivalent parameters are averaged into one row.

Vinyl fluoride — CH₂=CHF

6 atoms, 12 structural parameters. Published B₀ = 64,584.6, 10,636.9, 9,118.2 MHz; inertial defect +0.09 amu Å² (planar, as expected). The published geometry reproduces its own constants to 1.06%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1=C2	1.3290	1.3387	1.3102	1.3211	1.3212	1.3072
C1-F	1.3470	1.3571	1.3777	1.3444	1.3445	1.3721
C1-H4	1.0820	1.1023	1.0673	1.1233	1.1214	1.0669
C2-H5	1.0870	1.0842	1.0708	1.0880	1.0882	1.0701
C2-H6	1.0770	1.1198	1.0704	1.0984	1.0991	1.0696
F-C1=C2	120.80	116.66	121.22	121.90	121.89	120.86
H4-C1=C2	129.20	134.02	127.39	131.69	131.41	127.52
F-C1-H4	110.00	109.04	111.39	106.40	106.69	111.61
C1=C2-H5	119.00	123.08	121.51	120.02	119.92	121.33
C1=C2-H6	120.90	117.98	120.19	120.96	120.96	120.27
H5-C2-H6	120.10	118.85	118.30	119.01	119.10	118.40

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (18 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1=C2	1.3290	1.3387	1.3102	1.3196	1.3196	1.3205
C1-F	1.3470	1.3571	1.3777	1.3579	1.3579	1.3576
C1-H4	1.0820	1.1023	1.0673	1.0876	1.0874	1.0745
C2-H5	1.0870	1.0842	1.0708	1.0819	1.0819	1.0776
C2-H6	1.0770	1.1198	1.0704	1.0792	1.0791	1.0818
F-C1=C2	120.80	116.66	121.22	121.17	121.17	121.13
H4-C1=C2	129.20	134.02	127.39	132.48	132.47	128.57
F-C1-H4	110.00	109.04	111.39	106.32	106.36	110.29
C1=C2-H5	119.00	123.08	121.51	119.22	119.24	119.57
C1=C2-H6	120.90	117.98	120.19	121.17	121.19	120.78
H5-C2-H6	120.10	118.85	118.30	119.59	119.55	119.65

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

Acetyl fluoride — CH₃COF

7 atoms, 15 structural parameters. Published B₀ = 11,039.3, 9,685.7, 5,321.9 MHz; inertial defect -3.00 amu Å² (non-planar, as expected for a methyl group). The published geometry reproduces its own constants to 0.32%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5030	1.5313	1.4852	1.5114	1.5019	1.4812
C1=O	1.1810	1.1425	1.1876	1.1370	1.1572	1.1861
C1-F	1.3480	1.3994	1.3768	1.3842	1.3754	1.3725
C2-H5	1.0840	1.1192	1.0771	1.1299	1.1290	1.0763
C2-H6	1.0840	1.0715	1.0821	1.0607	1.0617	1.0792
O=C1-F	121.35	121.43	119.34	122.31	121.04	119.12
O=C1-C2	128.35	128.79	129.67	128.76	129.91	129.81
F-C1-C2	110.30	109.59	111.00	108.83	109.01	111.07
C1-C2-H5	110.40	109.11	109.83	109.63	110.21	110.06
C1-C2-H6	110.40	110.85	109.73	112.21	111.99	109.87

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (18 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5030	1.5313	1.4852	1.5068	1.5042	1.4840
C1=O	1.1810	1.1425	1.1876	1.1808	1.1812	1.1864
C1-F	1.3480	1.3994	1.3768	1.3507	1.3527	1.3701
C2-H5	1.0840	1.1192	1.0771	1.0807	1.0813	1.0797
C2-H6	1.0840	1.0715	1.0821	1.0745	1.0767	1.0795
O=C1-F	121.35	121.43	119.34	121.36	121.12	119.20
O=C1-C2	128.35	128.79	129.67	128.43	128.66	130.24
F-C1-C2	110.30	109.59	111.00	110.13	110.21	110.56
C1-C2-H5	110.40	109.11	109.83	110.68	110.60	110.25
C1-C2-H6	110.40	110.85	109.73	110.75	110.56	110.07

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

Fluoroethane — CH₃CH₂F

8 atoms, 18 structural parameters. Published $B_0 = 36,070.4, 9,364.6, 8,199.6$ MHz; inertial defect -6.34 amu Å² (non-planar, as expected for a methyl group). The published geometry reproduces its own constants to 0.64%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5050	1.5273	1.5091	1.5207	1.5133	1.5026
C1-F	1.3980	1.4053	1.4249	1.3972	1.4023	1.4184
C1-H4	1.0950	1.0893	1.0786	1.0989	1.1023	1.0809
C2-H6	1.0910	1.1330	1.0839	1.1146	1.1139	1.0838
C2-H7	1.0900	1.0738	1.0820	1.0762	1.0763	1.0841
F-C1-C2	109.70	106.03	108.83	109.33	109.49	109.03
F-C1-H4	106.04	105.66	106.62	105.18	105.52	106.47
C2-C1-H4	112.90	113.97	112.42	113.77	113.55	112.01
H4-C1-H5	108.81	110.64	109.60	108.81	108.53	110.54
C1-C2-H6	109.70	107.09	110.34	108.56	108.74	110.21
C1-C2-H7	110.40	110.24	110.58	109.05	109.08	110.41

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (18 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5050	1.5273	1.5091	1.5108	1.5096	1.5076
C1-F	1.3980	1.4053	1.4249	1.4023	1.4012	1.4062
C1-H4	1.0950	1.0893	1.0786	1.0991	1.0977	1.0893
C2-H6	1.0910	1.1330	1.0839	1.0937	1.0936	1.0972
C2-H7	1.0900	1.0738	1.0820	1.0873	1.0907	1.0883
F-C1-C2	109.70	106.03	108.83	109.62	109.65	109.41
F-C1-H4	106.04	105.66	106.62	106.07	106.41	106.89
C2-C1-H4	112.90	113.97	112.42	113.24	112.80	112.04
H4-C1-H5	108.81	110.64	109.60	108.17	108.41	109.31
C1-C2-H6	109.70	107.09	110.34	109.49	109.61	109.10
C1-C2-H7	110.40	110.24	110.58	109.47	110.18	110.64

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

7. What the numbers show

7.1 Theory has a consistent, one-directional C–F bias

Across the three molecules, theory puts the C–F bond +31, +29, +27 mÅ from the published value. Same sign, similar size, every time: at this level of theory the C–F bond is systematically too long by roughly 30 mÅ. That is not random error, it is a bias of the method, and no amount of tightening the geometry optimisation will remove it. It is exactly the kind of error experimental data is well placed to correct, and it is the single clearest argument for combining the two sources at all.

7.2 Spectroscopy alone is unreliable on thin data

With one measured species — three numbers against twelve to eighteen parameters — fitting to the data gives bond errors of 21, 35, 14 mÅ, starting from 22, 36, 23 mÅ. In two of the three the fit has barely improved on the deliberately wrong geometry it began with, moving it about 1 mÅ.

This is not the optimiser failing; it is the underdetermination showing through. Much of the damage lands on bonds involving hydrogen — the worst is off by +41, +46, +24 mÅ — because moving a hydrogen barely shifts the moments of inertia, so the fit will move one a long way to buy a small improvement. But it is not confined to hydrogens: in acetyl fluoride the C=O bond comes out 44 mÅ short and C–F 36 mÅ long. With three numbers and fifteen parameters there is nothing in the data forcing any particular answer, and the fit is entitled to any of them.

7.3 Spectroscopy alone is strong on full data

Once every symmetry-unique substitution is included the picture reverses. Bond errors fall to 7.3, 4.9, 4.1 mÅ, three to four times better than theory. This is the regime classical microwave structure determination has always operated in, and the result confirms the machinery uses the information correctly rather than discovering anything. Adding theory on top costs nothing and helps a little: the split hybrid gives 7.3, 4.1, 3.0 mÅ, equal or better in all three.

7.4 How the two sources are combined matters more than whether they are combined

The two hybrids are given identical inputs and differ only in the merging rule, yet they behave quite differently on thin data. The *split* rule hands every direction the data can see over to the data outright. On three observables that is too generous: some of those directions are seen only barely, and letting the data own them lets the same hydrogen-atom wandering through that spoils the experiment-only fit. The *joint* rule keeps theory's opinion in play everywhere, weighted by the distance over which theory is trusted, and that is what stops the drift.

On full data the ordering flips: there the prior is the thing that over-constrains, and the split rule is better. The reversal is clean — joint beats split in all three molecules on thin data, split beats joint in all three on full data — and the same reversal was found independently on fluorobenzene earlier in this project, so it is not a quirk of one system. The practical rule is simple: use the joint objective when measurements are scarcer than parameters, the split objective when they are not.

Angles are the one place the ordering is not uniform. In vinyl fluoride on full data the joint hybrid is far better on angles than anything else (0.43° against about 2° for the two data-led fits), because two of the angles around the double bond get distorted by the data-led fits and the prior holds them in place. In the other two molecules the split hybrid wins on angles as well. Angles are the weaker part of this comparison throughout, and worth treating with more caution than the bond lengths.

7.5 Does the hybrid beat theory?

The objective has to be chosen without knowing the answer, or the comparison is worthless. It can be: the rule in section 7.4 needs only the number of measurements and the number of parameters, both of which are known before any fitting happens. The table below applies that rule mechanically — joint below the parameter count, split at or above it — and compares the result with theory.

Molecule	Data level	Rule picks	theory	hybrid	change
Vinyl fluoride	parent only	joint	19.0	18.3	-0.7
Vinyl fluoride	all species	split	19.0	7.3	-11.7
Acetyl fluoride	parent only	joint	15.8	15.4	-0.4
Acetyl fluoride	all species	split	15.8	4.1	-11.7
Fluoroethane	parent only	joint	15.0	11.9	-3.1
Fluoroethane	all species	split	15.0	3.0	-12.0

RMS bond error in mÅ. The hybrid chosen by the counting rule beats theory in 6 of the 6 cases, and is closer on the C–F bond specifically in 6 of 6.

So: yes, with two qualifications. The margin is thin where data is scarce — under a milli-Ångström in two of the three molecules, because three numbers leave little to add — and becomes large, around 12 mÅ, only on full data. And the hybrid configured the other way is *worse* than plain theory in 2 of the 6 cases, so the objective is not a detail to leave at its default. The gain that is both large and consistent is the C–F bond, where a systematic error in the theory is exactly what data removes well.

8. Limitations

What	Why it matters
The substituted-species constants are derived, not measured.	Described in section 4. It makes the full-data level cleaner than reality and flatters the two data-using methods there. The parent-only level uses only published measured numbers and is unaffected.
Hartree–Fock with a small basis is a weak level of theory.	A better method would shrink theory's C–F bias and narrow the gap. That would change the size of the effect, not its direction; the point here is the comparison between ways of using a fixed quantum surface, not the absolute accuracy of any of them.
Comparing across structure definitions.	The published structures come from isotopic substitution; the quantum calculation produces the structure at the bottom of the energy well. These differ by a few mÅ through zero-point vibration, so part of every theory error quoted here is that definitional gap rather than a failure of the calculation.
Three molecules is a small sample.	The C–F bias is consistent across all three, and the objective-choice reversal reproduces a fourth system (fluorobenzene), but neither is established on a sample this size. They are indications worth acting on, not settled results.
One starting geometry per molecule.	Every method starts from the same displaced structure, which makes the comparison fair, but does not show how sensitive any of them is to where it starts.

9. Reproducing this

Both steps are in the repository and read no hidden state:

```
python scripts/monofluoro_benchmark.py
python scripts/build_monofluoro_report.py
```

Structures and constants live in dev/monofluoro_references.py, one entry per molecule with its citation. Every number in this report is read from the benchmark's JSON output rather than typed in, so the prose cannot drift from the run. The whole set takes about 7 minutes on one core.